

Zhuoli Ouyang | Curriculum Vitae

Beijing, China

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Education

Southern University of Science and Technology

Undergraduate, Information Engineering

GPA 3.88/4.00 Rank 6/48

- **Fields Elite Class (Honors)**, Department of Mathematics
- **Honors Class**, Department of Electronic & Electrical Engineering

Shenzhen, China

2023 to 2027 (expected)

2023 to 2024 (first year)

2024 to Present

Dartmouth College

Visiting Student (Research Only)

Advisor [Yaoqing Yang](#)

Hanover, NH

Mar 2025 to Oct 2025

Westlake University

Visiting Student (Research Only)

Advisor [Tailin Wu](#)

Hangzhou, China

Oct 2025 to Aug 2026

Research Experience

NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection

Submitted to ICLR 2027

Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu

Sole First Author. Independently led and completed all aspects of the research

Abstract NOVA solves moving-boundary PDEs using neural basis functions and boundary-compatible nullspace coordinates. Horizontal transport tracks the evolving boundary-compatible subspace, while physical-state reprojection handles rank transitions. Evaluation covers eight multiphysics benchmarks.

Contributions

- **Conceived the solver** by separating boundary enforcement from interior evolution through a boundary lift and nullspace coordinates.
- **Developed the theory and algorithms.** Derived horizontal nullspace transport, physical-state reprojection across rank changes, and a rank-saturation criterion for boundary sampling.
- **Wrote and organized all solver and benchmark code.** Ran all eight multiphysics benchmarks, baseline sweeps, and transport/sampling ablations, achieving the lowest boundary errors across all eight cases.
- **Led the literature review and wrote the full manuscript.**

OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States

Submitted to ICLR 2027

Zhuoli Ouyang, Yujun He, Hongbin Zhong, Han Wu, Haoning Jiang, Zhihai He

Sole First Author. Independently led and completed all aspects of the research

Abstract OrthVMS combines low-rank adaptive moments with an energy-preconditioned orthogonal gradient residual. The resulting full-rank updates use compact optimizer states, with convergence analysis under stated assumptions and empirical evaluation on LLaMA and GLA pretraining.

Contributions

- **Proposed the VMS coarse/fine optimizer and residual-energy preconditioner**, retaining full-space updates with one additional persistent vector per weight matrix.
- **Wrote and organized all optimizer and training code**, and developed convergence analysis under explicit assumptions on the joint update and projection-basis changes.
- **Designed and ran all LLaMA/C4 and GLA/FineWeb-Edu experiments** (60M to 1B). On GLA-1B, improved perplexity from 15.36 to 14.59 versus AdamW with approximately 32.5% less persistent model/optimizer-state memory.
- **Led the literature review and wrote the full manuscript.**

Research Experience (continued)

ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM

NeurIPS 2026

Fine-Tuning

Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu

Accept (Poster)

Sole First Author. Independently led and completed all aspects of the research

Abstract ENGINE applies Variational Multiscale (VMS) ideas to zeroth-order LLM fine-tuning. It couples low-rank optimization with full-space curvature feedback through a V-cycle, balancing gradient-estimation variance and curvature information to accelerate training.

Contributions

- **Originated the research idea** by adapting Variational Multiscale (VMS) theory from computational mechanics to LLM optimization, balancing variance and curvature in full-space and low-rank updates.
- **Developed and implemented the method.** Derived the VMS-EO correction and V-cycle updates, and wrote and organized the full codebase, including bidirectional transfer of parameters, momentum, and curvature.
- **Led all experiments.** Selected benchmarks, reproduced six zeroth-order baselines, and evaluated 16 model/task settings (1.5B to 13B parameters). Achieved $2.61\times$ forward-pass and $2.58\times$ wall-clock speedups over LOREN on LLaMA-3-8B/CB to match its best accuracy.
- **Wrote the full manuscript.** Conducted literature review and handled the full rebuttal process.

RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization ICML 2026

Shenyang Deng*, Zhuoli Ouyang*, Tianyu Pang, Zihang Liu, Ruochen Jin, Shuhua Yu, Yaoqing Yang

Accept (Regular)

Co-first Author (equal contribution), Lead for Experiments*

[\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

Abstract RMNP is a matrix optimizer for LLM pretraining that uses row-momentum normalization to compute an $O(mn)$ preconditioner. It provides efficient matrix updates, with empirical evaluation of preconditioning speed, perplexity, and training dynamics against AdamW and Muon.

Contributions

- **Wrote and organized the full optimizer and benchmark codebase**, and performed all experiments, including GPT/LLaMA pretraining comparisons with AdamW and Muon.
- **Measured $13\times$ to $44\times$ faster preconditioning** than Muon's Newton-Schulz step in GPT benchmarks (60M to 1.5B), alongside perplexity and training-dynamics comparisons.
- **Contributed to parts of the theory and manuscript.** Conducted literature review and participated throughout the rebuttal process.

Understanding Generalization from Embedding Dimension and Distributional Convergence ICML 2026

Junjie Yu*, Zhuoli Ouyang*, Haotian Deng*, Chen Wei, Wenxiao Ma, Jianyu Zhang, Zihan Deng, Quanying Liu

Accept (Regular)

Co-first Author (equal contribution), Lead for Theory*

[\[Paper\]](#)

Abstract For a fixed trained model, this work connects generalization to embedding intrinsic dimension and downstream sensitivity. Wasserstein convergence yields layer-wise risk bounds under regularity assumptions, with empirical evaluation across vision and language models.

Contributions

- **Established the theoretical framework** connecting embedding geometry, distributional convergence, and generalization. Conducted the literature review underpinning the theory.
- **Led the theoretical derivations.** Developed layer-wise risk bounds combining Wasserstein convergence, Lipschitz sensitivity, and representation approximation error, together with the final-layer specialization.
- **Guided experimental design** to test convergence scaling and the joint effects of embedding dimension and downstream sensitivity. Prepared rebuttal responses.

Honors & Awards

2026 ALT 2026 Best Student Paper Award

2026 ISIT 2026 Strong Accept

2026 DATE 2026 Oral Paper (ACEMARL)

2024 SUSTech, Department of Electronics and Electrical Engineering, Honor Class

2023 SUSTech, Mathematics Department, Fields Elite Class (Top 1%, Honor Class)

Other Skills

Programming Python, C, C++, MATLAB, LaTeX, etc.

Software PyTorch, PETSc, MATLAB, Fluent

Soccer Second-class National Athlete in Football

Piano Level 8, Central Conservatory of Music, Beijing

Hobbies Exploring mathematics, Coding, Philosophy and History, Playing piano, Playing soccer